HHCKLA Buddhist Leung Chik Wai College

6A error analysis (2026)

Student No.: 16 Name: _____ Marks: / 76

Class: 6A Date: _____ Marked date: _____

What have you learned in this chapter? _____

What can you do to improve your work next time? _____

1. Factorize

   (a) $5x^2y + 17xy - 12y$,

   (b) $5x^2y + 17xy - 12y - 10x + 6$.

(4 marks) (DSE-2026 PI Q04)

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2. The frequency distribution table and the cumulative frequency distribution table below show the distribution of the weights of 20 apples.

Weight (grams)	Frequency
91 – 100	2
101 – 110	a
111 – 120	6
121 – 130	7
131 – 140	h

Weight less than (grams)	Cumulative frequency
100.5	2
110.5	6
120.5	12
130.5	b
140.5	k

(a) Find a and b.

(b) Find the mean of the distribution.

(5 marks) (DSE-2026 PI Q09)

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3. The stem-and-leaf diagram below shows the distribution of the numbers of driving hours of a group of taxi drivers in a week.

Stem (tens)	Leaf (units)
2	5 6 9
3	2 3 3 6 6 7 8
4	0 0 2 2 5 5 7 7 7
5	w

The ratio of the range to the inter-quartile range of the distribution is 7 : 3.

(a) Find w.

(b) (i) Write down the mode of the distribution.

(ii) If two taxi drivers are randomly selected from the group, find the probability that the mean of their numbers of driving hours in the week exceeds the mode of the distribution.

(6 marks) (DSE-2026 PI Q10)

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4. When the polynomial $g(x)$ is divided by x^2+2x+a, the quotient and the remainder are $x+13$ and $b(x-3)$ respectively, where a and b are constants. It is given that $g(x)$ is divisible by $x-3$.

(a) Find a.

(b) Is there only one possible value of b such that at least two roots of the equation $g(x)=0$ are equal? Explain your answer.

(8 marks) (DSE-2026 PI Q14)

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5. It is given that P, Q and R are three points in a rectangular coordinate plane. The coordinates of P and Q are $(7,8)$ and $(-13,20)$ respectively. The equation of the perpendicular bisector of PR is $3x+5y-27=0$. Let C be the circle which passes through P, Q and R.

(a) Find the equation of C.

(b) The tangent to C at P and the tangent to C at R intersect at the point H. Is $\triangle HPR$ a right-angled triangle? Explain your answer.

(6 marks) (DSE-2026 PI Q16)

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6. In Figure 2, the base of the pyramid $VABCD$ is the quadrilateral $ABCD$. The face VAB of the pyramid lies on the horizontal ground. It is given that $AB = 90$ cm, $AD = 70$ cm, $CD = 100$ cm, $\angle ADC = 55°$ and $\angle BCD = 130°$.

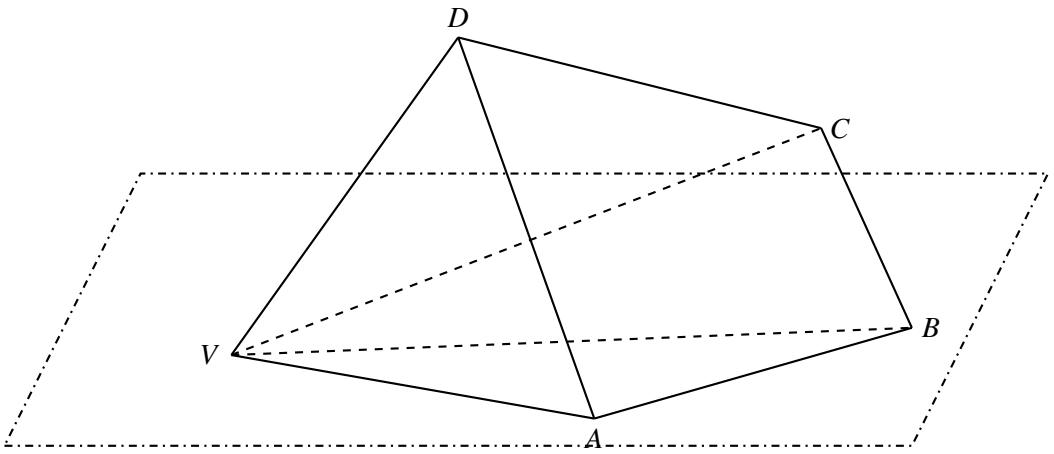


Figure 2

(a) Find

(i) the distance between A and C,

(ii) $\angle BAD$.

(b) It is given that $VD = 80$ cm. A student finds that the angle between the plane $ABCD$ and the horizontal ground is 65°. The student claims that the angle between VD and the horizontal ground is less than 50°. Is the claim correct? Explain your answer.

(6 marks) (DSE-2026 PI Q17)

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7. Let $G(n)$ be the nth term of a geometric sequence. It is given that $G(2) = 12$ and $G(5) = 768$.

(a) Find $G(1)$.

(b) Define $T(n) = \log_2 G(n)$ for all positive integers n. Find

(i) $T(n+1) - T(n)$,

(ii) the greatest value of k such that $T(1) + T(2) + T(3) + \cdots + T(k) < 10^6$.

(7 marks) (DSE-2026 PI Q18)

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8. Let $h(x) = -x^2 + 30kx + 4x - 225k^2 - 40k + 3$, where k is a positive constant. On the same rectangular coordinate system, denote the vertex of the graph of $y = h(x)$ and the vertex of the graph of $y = 14 - h(4-x)$ by A and B respectively.

(a) Using the method of completing the square, express, in terms of k, the coordinates of A.

(b) Write down, in terms of k, the coordinates of B.

(c) The coordinates of the point C are $(2 + 48k,\ 7 - 36k)$. Let I and J be the in-centre and the circumcentre of $\triangle ABC$ respectively.

(i) Let M be the mid-point of AB. Describe the geometric relationship between CM and AB. Explain your answer.

(ii) Express, in terms of k, the distance between I and J.

(iii) The coordinates of the point D are $(6 - 22k,\ 6 - 15k)$. Is it possible that $BDIJ$ is a parallelogram? Explain your answer.

(12 marks) (DSE-2026 PI Q19)

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9. The perimeter and the area of a sector are k cm and $3k$ cm^2 respectively. If the radius of the sector is 14 cm, then the arc length of the sector is

A. 19 cm.

B. 21 cm.

C. 28 cm.

D. 49 cm.

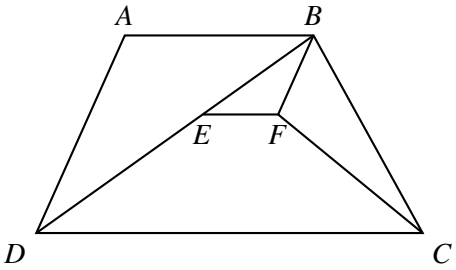
(DSE-2026 PII Q16)

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10. In the figure, $ABCD$ is a trapezium with $AB//DC$ and $AB : CD = 3 : 5$. Let E be a point lying on BD such that $BE : ED = 2 : 3$. It is given that $AB//EF$ and $AD//BF$. If the area of $\triangle ABD$ is 225 cm^2, then the area of the trapezium $CDEF$ is



A. 222 cm^2.

B. 260 cm^2.

C. 279 cm^2.

D. 285 cm^2.

(DSE-2026 PII Q18)

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11. It is given that $\triangle PQR$ is a right-angled triangle with $\angle QPR = 90^\circ$. Denote the area of $\triangle PQR$ by S. If $QR = p$, $PR = q$ and $\angle PQR = \theta$, which of the following must be true?

A. $S = \frac{1}{2}pq$

B. $S = \frac{1}{2}pq\sin\theta$

C. $S = \frac{1}{2}pq\cos\theta$

D. $S = \frac{1}{2}pq\tan\theta$

(DSE-2026 PII Q19)

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12. It is given that $ABCDEF$ is a regular 6-sided polygon. If AC and BF intersect at the point G, which of the following are true?

I. $BG//CE$

II. $\triangle ABG \sim \triangle BDC$

III. $\triangle AGF \cong \triangle BGC$

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

(DSE-2026 PII Q23)

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13. Which of the following statements about the graph of $y = 2 - \log_3(x+4)$ is/are true?

I. y increases when x increases.

II. The x-intercept of the graph is 5.

III. The y-intercept of the graph is 2.

A. I only

B. II only

C. I and III only

D. II and III only

(DSE-2026 PII Q35)

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14. It is given that $\log y$ is a linear function of x. The intercepts on the vertical axis and on the horizontal axis of the graph of the linear function are 2 and 1 respectively. If $y = ab^x$, find a.

A. 0.01

B. 0.1

C. 10

D. 100

(DSE-2026 PII Q36)

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15. Consider the following system of inequalities:

$$\begin{cases} y \leq 7 \\ 3x - 4y \leq 2 \\ 6x + 13y \geq 25 \end{cases}$$

Let R be the region which represents the solution of the above system of inequalities. If (x, y) is a point lying in R, then the least value of $5x + 4y + 28$ is

A. 1.

B. 19.

C. 42.

D. 106.

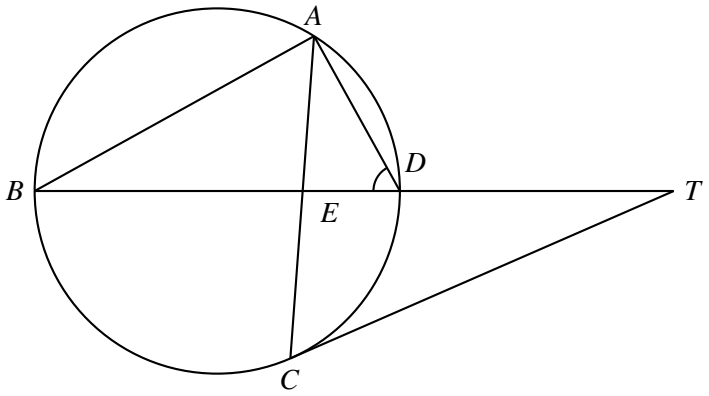
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16. In the figure, $ABCD$ is a circle. AC and BD intersect at the point E. BD is produced to the point T such that TC is tangent to the circle. It is given that $AB//TC$. If $\angle AED = 85°$ and $\angle CTE = 23°$, then $\angle ADE =$



A. 39°.

B. 56°.

C. 62°.

D. 72°.

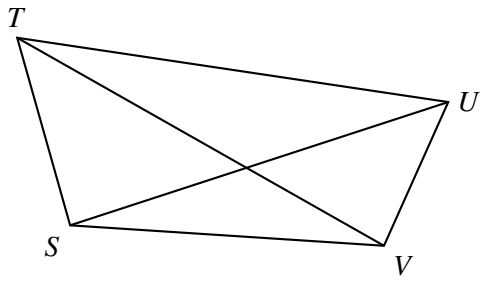
(DSE-2026 PII Q38)

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17. The figure shows the quadrilateral $STUV$, where $ST = 8$ cm and $SV = 5$ cm. If $\angle UTV = 30^\circ$ and $\angle TSU = \angle TVU = 90^\circ$, find $\angle SUV$ correct to the nearest degree.



A. 22°

B. 28°

C. 30°

D. 38°

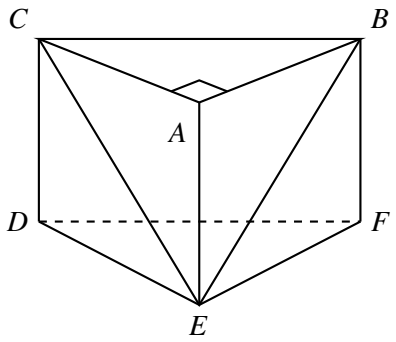
(DSE-2026 PII Q39)

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18. The figure shows the right prism $ABCDEF$, where $AB : AC : AE = 1 : 1 : 2$. Let θ be the angle between the plane ABC and the plane BCE. If $\angle BAC = 90°$, find $\sin\theta$.



A. $\frac{1}{3}$

B. $\frac{2}{3}$

C. $\frac{\sqrt{2}}{4}$

D. $\frac{2\sqrt{2}}{3}$

(DSE-2026 PII Q40)

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19. For $90^\circ \leq \theta \leq 360^\circ$, how many roots does the equation $6\sin^4\theta - 5\sin^2\theta + 1 = 0$ have?

A. 2

B. 4

C. 6

D. 8

(DSE-2026 PII Q41)

I can finish _____ questions

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